

scTHREAD: a single-cell long-read transcriptome atlas

<https://scthread.ai4sc.ac.cn>

Biological coverage

Tissues · disease · development · two species



453 runs · 34 datasets · 923,389 cells

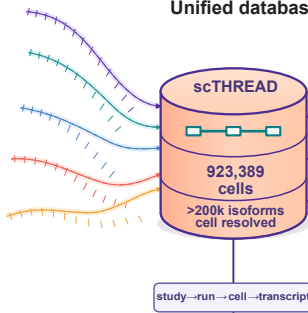
Human + mouse · ONT + PacBio



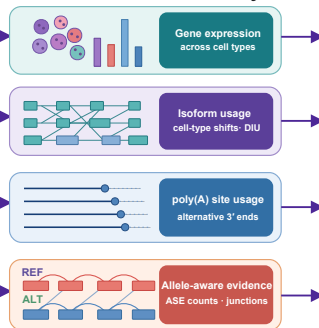
Registry-backed tissues & contexts · cells

Frontal cortex	121.2k	CCUS marrow	117.3k
iNeuron trajectory	74.5k	Cross-platform BM mix	71.8k
Mouse gastrulation	68.4k	Myeloma marrow	68.4k
Splicing dynamics	55.5k	Mouse testis	39.6k
iPSC to cortical neuron	35.0k	Human ovary	31.8k

Unified database



Four linked RNA evidence layers



Explore genes & cell types

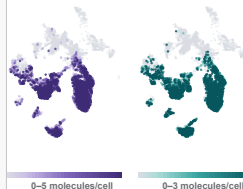
PTPRC / CD45

PTPRC isoform expression

same 71,913-cell embedding

ENST...67364

ENST...97630



all positive cells shown
+ stratified double-zero background

Open data access

Browse Search Download

CSV · JSON · API

PTPRC isoform usage (% of all 23)

ENST...67364	19	42	29	59	24	59	17	44
ENST...97630	34	22	27	14	32	18	40	17
	B	DC	Ery	Mo	NK	PI	Pro	T

Precomputed gene-level annotations

significant / tested genes · FDR or $q < 0.05$ + effect size

DIU 2,008 / 8,092 genes

APA 2,558 / 10,531 genes

ASE 9,880 / 37,415 genes

DIU · APA cell-type-differential (effect ≥ 0.20) · ASE allelic imbalance ($|dev| \geq 0.10$)

Export · CSV · JSON · API

Human & mouse tissues · cell types · isoforms · poly(A) sites · allele-aware expression